

EXECUTIVE SUMMARY

Traffic related sources are a significant contributor of particulate matter, particularly in urban environments and major cities. Traffic related particles can be distinguished into: **exhaust traffic related particles**, which are emitted as a result of incomplete fuel combustion and lubricant volatilization during the combustion procedure, and **non-exhaust traffic related particles**, which are either generated from non-exhaust traffic related sources such as brake, tyre, clutch and road surface wear or already exist in the environment as deposited material and become resuspended due to traffic induced turbulence. It is estimated that exhaust and non-exhaust sources contribute almost equally to total traffic-related PM₁₀ emissions. However, as exhaust emissions control become stricter, relative contributions of non-exhaust sources to traffic related emissions will increasingly become more significant. The aim of the present literature review study is to present the state-of-the-art of the different aspects regarding particulate emissions resulting from non-exhaust sources and particularly from brake and tyre wear. For this reason several different literature sources such as peer reviewed papers, research project reports, technical publications, as well as licentiate and doctoral theses were examined and the most significant findings in terms of importance, physicochemical characteristics, EFs and possible adverse health effects are discussed. Brake wear contributes 16-55% by mass to total non-exhaust traffic related PM₁₀ emissions in urban environments, while in freeways this contribution is significantly lower (~3% by mass) due to lower braking frequency. Somewhat lower contributions (5-30% by mass) have been reported for tyre wear. Most researchers report unimodal mass size distributions of brake wear PM₁₀, with a mass weighed mean diameter of 2-6 µm. On the other hand, tyre wear PM₁₀ often displays a rather bimodal distribution with one peak lying within the fine particle size range and the other within the coarse range (5-9 µm). It is estimated that almost 40-50% by mass of generated brake wear particles and 0.1-10% by mass of tyre wear particles is emitted as PM₁₀. The rest may be attracted